The Double Pendulum

Elizabeth Kamminga and Maddie Moffett

PHSX 333 — Project 1

1 Introduction

The double pendulum is one example of a chaotic physical system. It consists of two rods of equal length ℓ and two equal masses m connected end to end– the second pendulum hangs from the bob of the first, which itself hangs from a fixed pivot.

The motion of the system is governed by four coupled, nonlinear ordinary differential equations (ODEs) describing the angular positions θ_1, θ_2 and angular velocities ω_1, ω_2 of the two arms:

$$\frac{d\theta_1}{dt} = \omega_1, \qquad \frac{d\theta_2}{dt} = \omega_2 \tag{1}$$

$$\frac{d\omega_1}{dt} = \frac{-\omega_1^2 \sin(2\theta_1 - 2\theta_2) + 2\omega_2^2 \sin(\theta_1 - \theta_2) + \frac{g}{\ell}\left[\sin(\theta_1 - 2\theta_2) + 3\sin\theta_1\right]}{3 - \cos(2\theta_1 - 2\theta_2)} \tag{2}$$

$$\frac{d\omega_2}{dt} = \frac{4\omega_1^2 \sin(\theta_1 - \theta_2) + \omega_2^2 \sin(2\theta_1 - 2\theta_2) + 2\frac{g}{\ell}\left[\sin(2\theta_1 - \theta_2) - \sin\theta_2\right]}{3 - \cos(2\theta_1 - 2\theta_2)} \tag{3}$$

These equations are nonlinear because they involve trigonometric functions of the angles and products of the angular velocity terms. The nonlinearity is what gives the system its chaotic character: for large initial angles, initial conditions diverge exponentially over time, making long-term prediction impossible. By contrast, for small initial angles the nonlinear coupling terms become negligible, and the system reduces to two weakly coupled linear pendulums exhibiting regular, predictable motion.

Because no analytical solution exists for the general double pendulum, the equations of motion must be integrated numerically. A key part for the accuracy of any numerical integration scheme is conservation of energy. Since the system is conservative, the total mechanical energy $E = T + V$ must remain constant throughout the simulation. Any drift in energy over time is therefore a direct measure of numerical error.

This project investigates the dynamics of the double pendulum using a fourth-order Runge-

Kutta (RK4) integration algorithm implemented in Python. The goals are to derive and verify total energy conservation, characterize the accuracy of the numerical scheme as a function of time-step size, and explore how initial conditions determine whether the system exhibits regular or chaotic behavior.

2 Numerical Methods

2.1 Derivation of Total Energy

The Cartesian coordinates of the two masses follow directly from the geometry of the double pendulum:

$$x_1 = \ell \sin\theta_1, \qquad y_1 = -\ell\cos\theta_1 \tag{4}$$

$$x_2 = \ell \sin\theta_1 + \ell\sin\theta_2, \qquad y_2 = -\ell\cos\theta_1 - \ell\cos\theta_2 \tag{5}$$

Taking time derivatives yields the velocity components:

$$\dot{x}_1 = \ell\omega_1\cos\theta_1, \qquad \dot{y}_1 = \ell\omega_1\sin\theta_1 \tag{6}$$

$$\dot{x}_2 = \ell\omega_1\cos\theta_1 + \ell\omega_2\cos\theta_2, \qquad \dot{y}_2 = \ell\omega_1\sin\theta_1 + \ell\omega_2\sin\theta_2 \tag{7}$$

The total kinetic energy is $T = \frac{1}{2}m(\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2}m(\dot{x}_2^2 + \dot{y}_2^2)$. Substituting and simplifying using $\cos(\theta_1 - \theta_2) = \cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2$ gives:

$$T = \frac{1}{2}m\ell^2\left[2\omega_1^2 + \omega_2^2 + 2\omega_1\omega_2\cos(\theta_1 - \theta_2)\right] \tag{8}$$

The potential energy is taken relative to the pivot point, with $y = 0$ at the pivot:

$$V = mgy_1 + mgy_2 = -mg\ell(2\cos\theta_1 + \cos\theta_2) \tag{9}$$

The total mechanical energy of the system is therefore:

$$\boxed{E = \frac{1}{2}m\ell^2\left[2\omega_1^2 + \omega_2^2 + 2\omega_1\omega_2\cos(\theta_1 - \theta_2)\right] - mg\ell(2\cos\theta_1 + \cos\theta_2)} \tag{10}$$

2.2 Fourth-Order Runge-Kutta Integration

To solve the four coupled equations of motion, an RK4 algorithm was implemented manually in Python. The algorithm was coded from scratch rather than using a library routine in order to maintain full control over the integration process.

The state of the system at any given time is represented as a vector of four quantities: $[\theta_1,\ \theta_2,\ \omega_1,\ \omega_2]$. A derivative function was written to evaluate the time derivatives of this state vector at any moment, returning $[\dot{\theta}_1,\ \dot{\theta}_2,\ \dot{\omega}_1,\ \dot{\omega}_2]$ directly from the equations of motion.

The RK4 algorithm advances the state forward by one time step h by computing four slope estimates:

$$k_1 = h \cdot f(r_n) \tag{11}$$

$$k_2 = h \cdot f\left(r_n + \tfrac{1}{2}k_1\right) \tag{12}$$

$$k_3 = h \cdot f\left(r_n + \tfrac{1}{2}k_2\right) \tag{13}$$

$$k_4 = h \cdot f(r_n + k_3) \tag{14}$$

The new state is then computed as the weighted average:

$$r_{n+1} = r_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{15}$$

The midpoint estimates k_2 and k_3 receive twice the weight of the endpoint estimates k_1 and k_4, which is what gives RK4 its fourth-order accuracy. Two small helper functions were written to perform scalar multiplication, allowing the k values to be computed and combined cleanly.

The integration proceeds by stepping forward from $t = 0$ to $t = 100$ s in increments of h, storing θ_1, θ_2, ω_1, and ω_2 at each time step. From these stored values, three quantities are computed and plotted: the total mechanical energy, the x-positions of both masses, and the

y-positions of both masses. To reduce plotting overhead, only every tenth stored data point is used when generating figures.

3 Results

3.1 Energy Conservation and Time-Step Selection

The simulation was run using $g = 9.80$ m/s^2, $\ell = 0.40$ m, $m = 1$ kg, and initial conditions $\theta_1 = \theta_2 = \pi/2$ with $\omega_1 = \omega_2 = 0$, integrated from $t = 0$ to $t = 100$ s.

Different values of h were tested to determine how small a time step is required to keep the energy loss below 10^{-5} J. The results are summarized in Table 1.

Time Step h (s)	Energy Loss after 100 s (J)
0.01	$\gg 10^{-5}$
0.002	$\approx 1 \times 10^{-5}$
0.001	$\approx 3 \times 10^{-7}$

Table 1: Energy loss after 100 s of integration for different time steps.

A time step of $h = 0.002$ s is sufficient to meet the 10^{-5} J threshold. However, $h = 0.001$ s was ultimately chosen for all simulations in order to obtain a more accurate result, achieving an energy loss of approximately 3×10^{-7} J — more than an order of magnitude better than required. The total energy plot at this time step is nearly flat across the full 100 s integration window, confirming that the numerical solution is well-converged. See Figures 1–6 at the end of this document.

The x- and y-positional plots for $h = 0.001$ s show highly irregular, aperiodic oscillations for both masses, consistent with the chaotic nature of the system at these large initial angles.

3.2 Systematic Variation of Initial Conditions

To explore the transition between regular and chaotic behavior, the simulation was run for all 25 combinations of initial angles drawn from the set $\{\pi/6,\ \pi/4,\ \pi/3,\ \pi/2,\ \pi\}$, with initial velocities held at zero throughout. All other parameters were kept identical to those in Section 3.1.

Chaotic behavior was observed in 13 of the 25 combinations — specifically, all cases in which at least one of the two initial angles was $\pi/2$ or π. In these cases, the plots are highly irregular, with large-amplitude, aperiodic oscillations that show no repeating pattern. The second mass in particular traces an erratic path, driven by strong nonlinear coupling from the large-angle first arm. These combinations are documented in Appendix A.

Regular behavior was observed in 12 of the 25 combinations — those in which both initial angles were drawn from $\{\pi/6,\ \pi/4,\ \pi/3\}$. In these cases, the positional plots show smooth, nearly periodic oscillations. When both angles are small ($\pi/6$ and $\pi/4$), the motion closely resembles that of two weakly coupled linear pendulums, with the nonlinear coupling terms contributing only a small correction. These combinations are documented in Appendix B.

The transition toward chaos is progressive rather than abrupt. Cases such as $(\theta_1, \theta_2) = (\pi/3,\ \pi/2)$ show behavior that is more complex than the small-angle cases but not as fully erratic as the π-angle cases, suggesting a slower onset of chaos as the initial energy of the system increases. This is consistent with the physical expectation that the nonlinear terms become increasingly dominant as the amplitudes grow.

4 Conclusion

This project successfully modeled the dynamics of a double pendulum using a manually implemented fourth-order Runge-Kutta algorithm in Python. The total mechanical energy of the system was derived analytically and used throughout as a diagnostic for numerical accuracy.

For the initial conditions $\theta_1 = \theta_2 = \pi/2$ and $\omega_1 = \omega_2 = 0$, a time step of $h = 0.001$ s was found to yield an energy loss of approximately 3×10^{-7} J over 100 s of integration, well below the 10^{-5} J threshold. A coarser time step of $h = 0.002$ s was also found to be sufficient, though the finer step was retained for greater accuracy.

Systematic variation of initial angles confirmed the sensitivity to initial conditions that characterizes chaotic systems. When both initial angles were at most $\pi/3$, the system displayed smooth, nearly periodic motion consistent with the linear pendulum approximation. When at least one initial angle was $\pi/2$ or π, the system exhibited chaotic behavior with large-amplitude, aperiodic oscillations. This transition reflects the physical reality that the non-linear coupling terms in the equations of motion become dominant at large displacements.

A Chaotic Initial Condition Combinations

The following initial angle combinations (with $\omega_1 = \omega_2 = 0$) produced chaotic motion. In each case, at least one angle is $\pi/2$ or π.

$$(\pi,\ \pi/6),\quad (\pi,\ \pi/4),\quad (\pi,\ \pi/3),\quad (\pi,\ \pi/2),\quad (\pi,\ \pi)$$

$$(\pi/6,\ \pi),\quad (\pi/4,\ \pi),\quad (\pi/3,\ \pi)$$

$$(\pi/2,\ \pi/6),\quad (\pi/2,\ \pi/4),\quad (\pi/2,\ \pi/3),\quad (\pi/2,\ \pi/2),\quad (\pi/2,\ \pi)$$

B Regular Initial Condition Combinations

The following initial angle combinations (with $\omega_1 = \omega_2 = 0$) produced regular, approximately periodic motion consistent with weakly coupled linear pendulums.

$$(\pi/6,\ \pi/6),\quad (\pi/6,\ \pi/4),\quad (\pi/6,\ \pi/3),\quad (\pi/6,\ \pi/2)$$

$$(\pi/4,\ \pi/6),\quad (\pi/4,\ \pi/4),\quad (\pi/4,\ \pi/3),\quad (\pi/4,\ \pi/2)$$

$$(\pi/3,\ \pi/6),\quad (\pi/3,\ \pi/4),\quad (\pi/3,\ \pi/3),\quad (\pi/3,\ \pi/2)$$

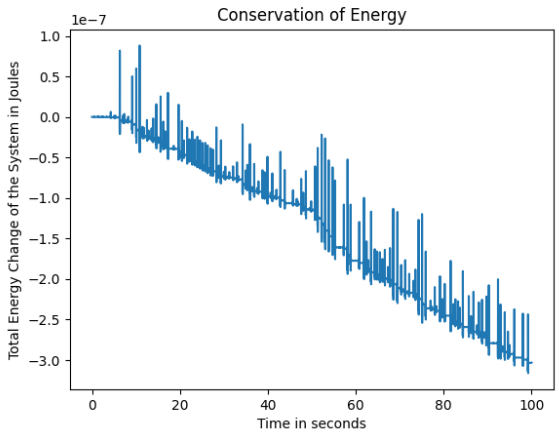

Figure 1: Total energy as a function of time for $h = 0.001$ s.

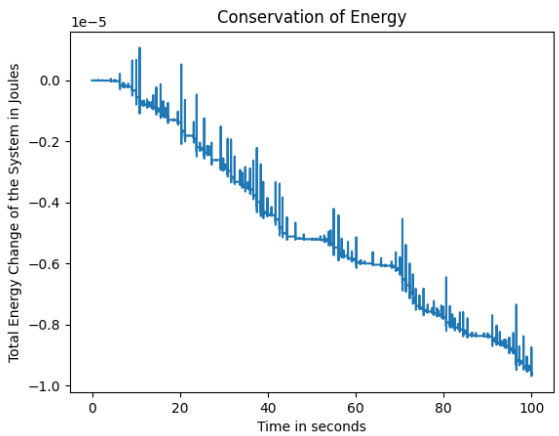

Figure 2: Total energy as a function of time for $h = 0.002$ s.

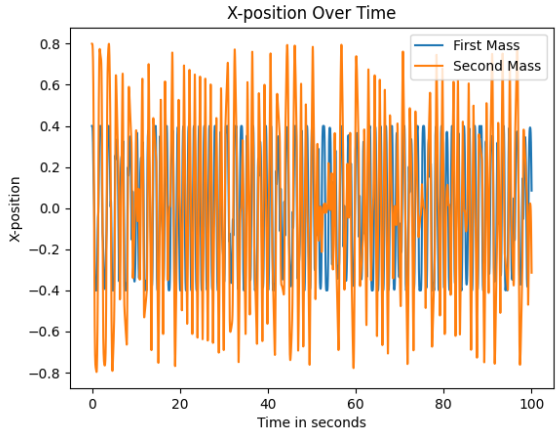

Figure 3: x-positions of both masses as a function of time for $h = 0.001$ s.

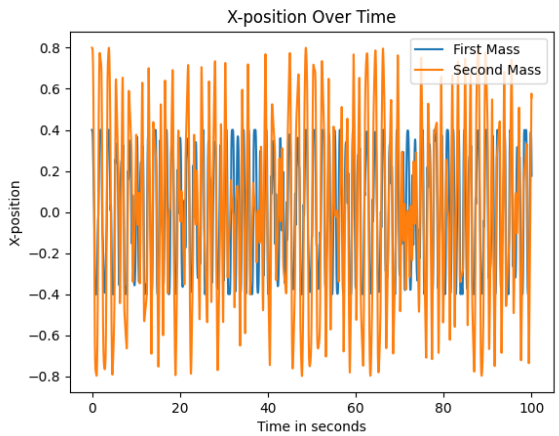

Figure 4: x-positions of both masses as a function of time for $h = 0.002$ s.

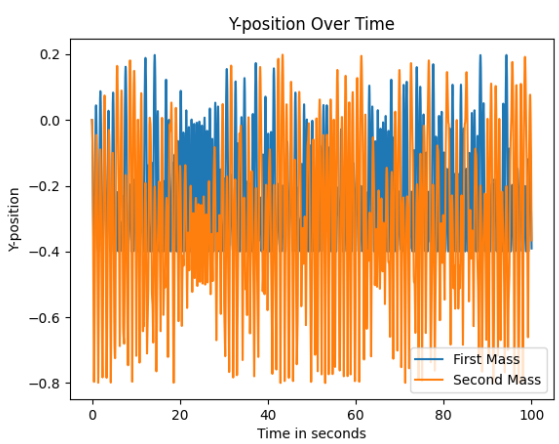

Figure 5: y-positions of both masses as a function of time for $h = 0.001$ s.

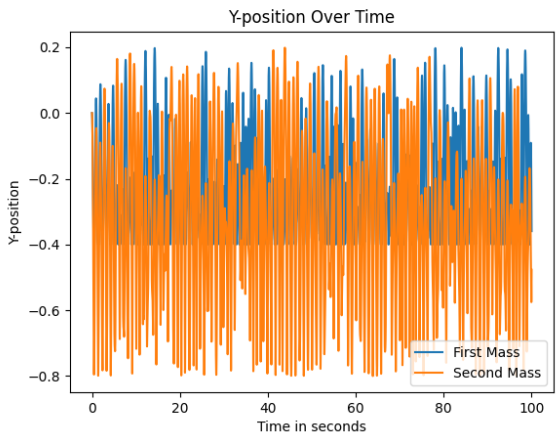

Figure 6: y-positions of both masses as a function of time for $h = 0.002$ s.